

1. Parts of a Program & `cout` Object

Write a C++ program that prints the following three lines exactly as shown:

```
text
Welcome to C++!
Let's learn the basics.
This is my first program.
```

Use only **one** `cout` statement (with multiple `<<` operators) to produce this output.

2. The `#include` Directive

Write a program that includes the `<iostream>` header and prints the sentence:
The `iostream` header is included.

3. Identifiers & Integer Data Types

Write a program that does the following:

- Declare an **integer** variable named `studentCount`.
- Assign the value `25` to it.
- Print the value with an appropriate message, e.g.,
`Number of students: 25`

4. Variable, Literals and Assignment Statement

Write a program that:

- Creates an `int` variable called `age` and initializes it with the literal `18` (using the assignment operator).
- Prints the value.
- Changes the value of `age` to `20` using another assignment.
- Prints the new value.

5. The `char` Data Type

Write a program that:

- Declares a `char` variable named `letter`.
- Assigns the character 'B' to it.
- Prints the character along with a message, for example:
`The chosen letter is: B`

6. The `bool` Data Type

Write a program that:

- Declares a `bool` variable called `isRaining` and initializes it to `false`.
- Prints the value of `isRaining`.
(It should print `0`, which represents `false`.)
- Changes the variable to `true` and prints it again.
(It should print `1`.)

7. Floating-Point Data Types

Write a program that:

- Declares a `float` variable named `temperature` and initializes it with `36.6`.
- Declares a `double` variable named `pi` and initializes it with `3.1415926535`.
- Prints both values with descriptive labels.

8. The `C++ string` Class

Write a program that:

- Includes the `<string>` header.
- Declares a `std::string` variable called `name`.
- Assigns your full name to it (for example, "Alex Johnson").
- Prints a greeting like: `Hello, Alex Johnson!`

9. Determining the Size of a Data Type (`sizeof`)

Write a program that uses the `sizeof` operator to print the size (in bytes) of the following data types:

`int`, `char`, `float`, `double`, and `bool`.

Each result should be on a separate line with a label, for example:

Size of `int`: 4

10. More about Variable Assignments & Initialization

Write a program that:

- Declares an `int` variable `x` and initializes it in the same statement to `10`.
- Declares a second `int` variable `y` (uninitialized), then assigns it the value `x + 5` on the next line.
- Prints both `x` and `y`.

11. Arithmetic Operators

Write a program that declares two integer variables `a = 15` and `b = 4`.

Calculate and print the results of the following operations (each on a new line):

- Addition (`a + b`)
- Subtraction (`a - b`)
- Multiplication (`a * b`)
- Division (`a / b`) – notice what happens with integer division
- Modulus (`a % b`) – remainder of the division

12. Scope

Write a program that:

- Declares an integer variable `value = 100` inside `main()`.
- Prints `value`.
- Opens a new block (a pair of curly braces `{}`).
- Inside that block, declares another integer variable also named `value` and initializes it to `50`.
- Prints the inner `value` from inside the block.

- After the block, prints the outer `value` again.
Observe which variable is printed each time.

13. Comments

Write a program that prints the message `"Comments are important."`.
In the same program:

- Add a **single-line comment** explaining what the `#include <iostream>` line does.
- Add a **multi-line comment** at the top of the file that includes your name, the date, and a brief description of the program.